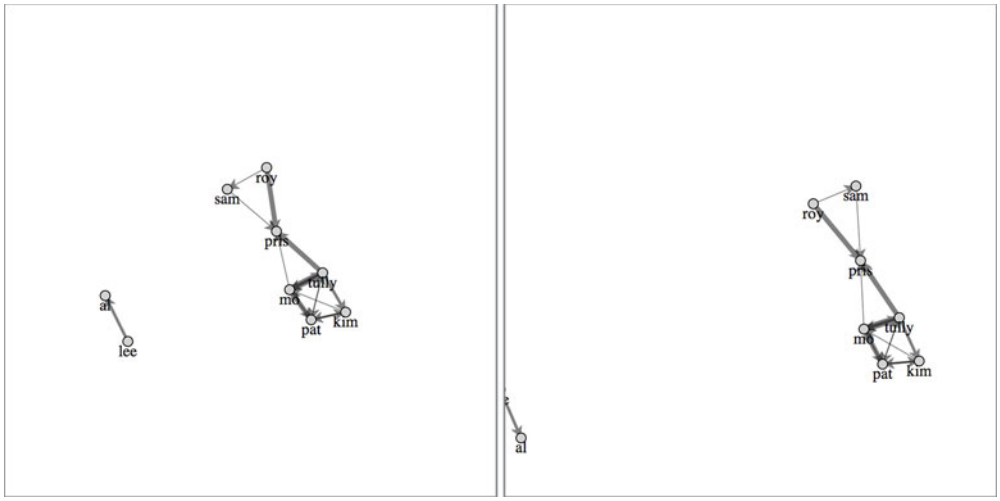


Figure 6.13 Increasing the gravity to .2 (left) pulls the two components closer to the center of the layout area. Decreasing the gravity to .05 (right) allows the small component to drift offscreen.

LINKDISTANCE

Attraction between nodes is determined by setting the `linkDistance` property, which is the optimal distance between connected nodes. One of the reasons we needed to set our charge so high was because the `linkDistance` defaults to 20. If we set it to 50, then we can reduce the charge to -100 and produce the results in figure 6.14.

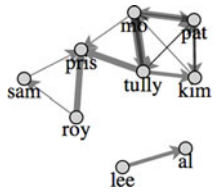


Figure 6.14 With `linkDistance` adjusted, our network becomes much more readable.

Setting your `linkDistance` parameter too high causes your network to fold back in on itself, which you can identify by the presence of prominent triangles in the network visualization. Figure 6.15 shows this folding occur with `linkDistance` set to 200.

You can set `linkDistance` to be a function and associate it with edge weight so that edges with higher or lower weight values have lower or higher distance settings. A better way to achieve that effect is to use `linkStrength`.

LINKSTRENGTH

A force layout is a physical simulation, meaning it uses physical metaphors to arrange the network to its optimal graphical shape. If your network has stronger and weaker

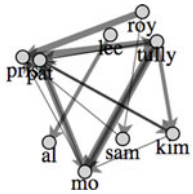


Figure 6.15 Distortion based on high `linkDistance` makes it look like Pris is connected to Pat and otherwise clusters nodes together despite their being unrelated.